

Exercises

1. The ability to adjust to changes in product mix, production volume, or design is _____.
 - (A) Flexibility
 - (B) Delivery Speed
 - (C) Customization
 - (D) Conformance quality
 - (E) More than one of the above is correct

2. Taco Bell processes on average 1500 customers per day (15 hours). On average, there are 75 customers in the joint (waiting to place orders, waiting for the order to arrive, eating, going back to the counter for another order, etc.) What is the average amount of time a customer spends at Taco Bell?
 - (A) $0 \leq X < 0.2$ hours
 - (B) $0.2 \text{ hours} \leq X < 0.4$ hours
 - (C) $0.4 \text{ hours} \leq X < 0.6$ hours
 - (D) $0.6 \text{ hours} \leq X < 0.8$ hours
 - (E) $0.8 \text{ hours} \leq X < 1$ hour
 - (F) $1 \text{ hour} \leq X < 1.2$ hours
 - (G) $1.2 \text{ hours} \leq X < 1.4$ hours
 - (H) $1.4 \text{ hours} \leq X < 1.6$ hours
 - (I) $1.6 \text{ hours} \leq X < 1.8$ hours
 - (J) $1.8 \text{ hours} \leq X < 2$ hours
 - (K) $X \geq 2$ hours

3. Motorola Inc. sells 300 million dollars worth of cellular equipment per year. The average amount in Accounts Receivable is 45 million dollars. On average, how much time does it take from a customer is billed to the time a payment is received?
 - (A) $0 \leq T < 0.2$ years
 - (B) $0.2 \text{ years} \leq T < 0.4$ years
 - (C) $0.4 \text{ years} \leq T < 0.6$ years
 - (D) $0.6 \text{ years} \leq T < 0.8$ years
 - (E) $0.8 \text{ years} \leq T < 1$ year
 - (F) $1 \text{ year} \leq T < 1.2$ years
 - (G) $1.2 \text{ years} \leq T < 1.4$ years
 - (H) $1.4 \text{ years} \leq T < 1.6$ years
 - (I) $1.6 \text{ years} \leq T < 1.8$ years

4. An automated pizza vending machine heats and dispenses a slice of pizza in 4 minutes. Customers arrive at an average rate of one every 6 minutes. On average, each customer spends about 8 minutes waiting in the line. What is the average number of customers at the vending machine?
- (A) $0 \leq Y < 2$
 (B) $2 \leq Y < 4$
 (C) $4 \leq Y < 6$
 (D) $6 \leq Y < 8$
 (E) $8 \leq Y < 10$
 (F) $10 \leq Y < 12$
 (G) $12 \leq Y < 14$
 (H) $14 \leq Y < 16$
 (I) $16 \leq Y < 18$
 (J) $18 \leq Y < 20$
 (K) $Y \geq 20$
5. Consider the baggage check-in of a small airline. Check-in data indicate that from 9 a.m. to 10 a.m., 255 passengers checked in. Moreover, based on counting the number of passengers waiting in line, airport management found that the average number of passengers waiting for check-in was 35. What is the average waiting time in line for a passenger, T?
- (A) $T < 2$ minutes
 (B) $2 \text{ minutes} \leq T < 4 \text{ minutes}$
 (C) $4 \text{ minutes} \leq T < 6 \text{ minutes}$
 (D) $6 \text{ minutes} \leq T < 8 \text{ minutes}$
 (E) $8 \text{ minutes} \leq T < 10 \text{ minutes}$
 (F) $10 \text{ minutes} \leq T < 12 \text{ minutes}$
 (G) $12 \text{ minutes} \leq T < 14 \text{ minutes}$
 (H) $14 \text{ minutes} \leq T < 16 \text{ minutes}$
 (I) $16 \text{ minutes} \leq T < 18 \text{ minutes}$
6. An assembly line is to operate eight hours per day with a desired output of 240 units per day. The following table contains information on this product's task times and precedence relationships:

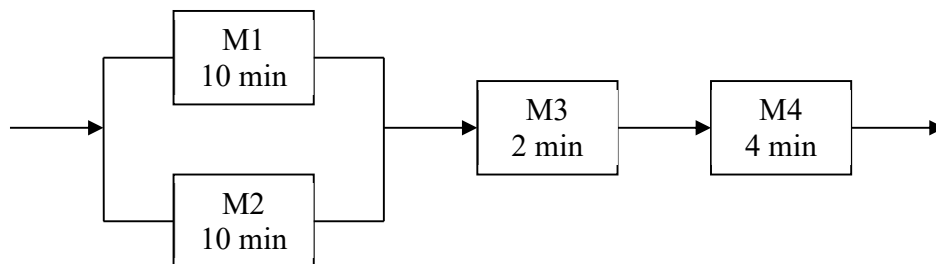
Task	Time (seconds)	Immediate Predecessor(s)
A	60	-
B	80	A
C	20	A
D	50	A
E	90	B, C
F	30	C, D
G	30	E, F
H	60	G

- (a) Draw and label a precedence diagram for the above tasks.
 (b) What is the cycle time for the line?
 (c) Balance the line using the longest task time rule.
 (d) What is the efficiency of your line balance in (c)?

7. UNT Motors, Inc. wishes to design a manufacturing assembly line that operates one shift (480 minutes) per day and the following tasks must be performed on the line:

Task	Time (seconds)	Immediate Predecessor(s)	Task	Time (seconds)	Immediate Predecessor(s)
A	55	-	F	60	B
B	30	A	G	40	F
C	40	A	H	25	E, D
D	100	C	I	60	G, H
E	45	C			

- (a) Draw and label a precedence diagram for the above tasks.
- (b) What is the maximum production rate per day for UNT Motors' production line?
- (c) Answer each of the following.
- What output will result if the cycle time is 3.0 minutes?
 - What cycle time would provide an output of 200 units per day?
- (d) UNT Motors, Inc. has forecasted a weekly demand of 4320 units and wants to design the assembly line to meet that demand. The company has decided to have three eight-hour shifts (480 minutes per shift) five days per week.
- Compute the cycle time.
 - Determine the theoretical minimum number of workstations to meet this demand.
- (e) Assume system requirements (output desired and operating parameters) presented in part d. Perform an initial balance of the assembly line for UNT Motors using the **longest task time rule**, Inc. Show the work stations, eligible tasks for assignment, tasks actually assigned, and idle times.
- (f) What is the efficiency of the actual line balance from above?
8. Suppose a process flow diagram is as follows. In step 1, the two machines (M1 & M2) are identical. Assume the process is in steady state and there is no buffer between each step. Suppose the average WIP inventory is 3.2 units. What is the average flow time of the process, X?



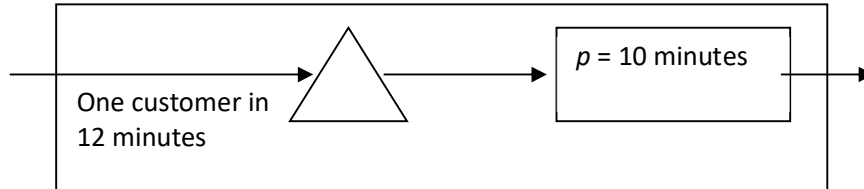
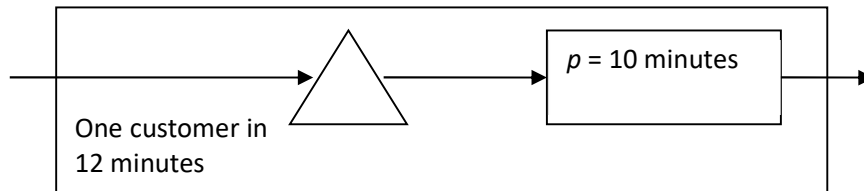
- (A) $X < 2$ minutes
- (B) $2 \text{ minutes} \leq X < 4 \text{ minutes}$
- (C) $4 \text{ minutes} \leq X < 6 \text{ minutes}$
- (D) $6 \text{ minutes} \leq X < 8 \text{ minutes}$
- (E) $8 \text{ minutes} \leq X < 10 \text{ minutes}$
- (F) $10 \text{ minutes} \leq X < 12 \text{ minutes}$
- (G) $12 \text{ minutes} \leq X < 14 \text{ minutes}$
- (H) $14 \text{ minutes} \leq X < 16 \text{ minutes}$
- (I) $16 \text{ minutes} \leq X < 18 \text{ minutes}$
- (J) $18 \text{ minutes} \leq X < 20 \text{ minutes}$
- (K) $X \geq 20 \text{ minutes}$

9. The layout of a process consists of two welding machines in parallel, one grinding and one drilling machine. Two panels are released simultaneously to two welding machines where some parts are fitted to the panels using these welding machines. The fitting operation requires 20 minutes per panel. The next step is grinding. Panels are placed in a buffer in front of the grinding machine if the machine is busy. The grinding operation takes 15 minutes per panel. The last step is drilling. After the grinding operation, panels are moved to a buffer in front of the drilling machine if the drilling machine is busy. The drilling operation takes 25 Minutes per panel.
- Draw a process flow diagram.
 - What's the hourly capacity of the process?
 - How long will it take to fill a rush order of one panel?
 - Suppose you now have a budget of \$6000 to buy additional machines to add to the process in order to increase your capacity. The unit prices for the welding, grinding, and drilling machines are all the same, i.e., \$2000 each. How are you going to spend the money? Please explain your reasons and calculate the new capacity after investment.
10. A manufacturer of fiberglass fittings is planning to add a new line of smaller fittings, and you have been asked to develop a balanced line given the following task times and precedence relationships:

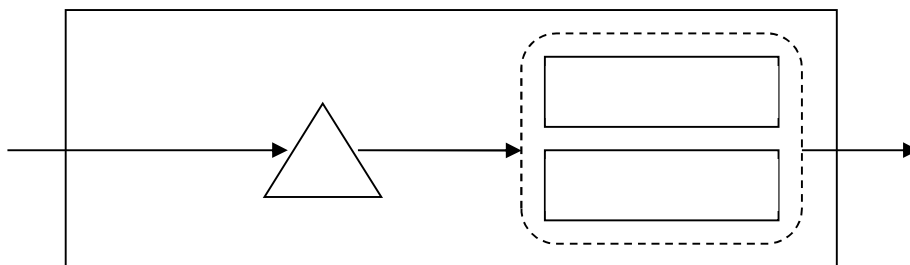
Task	Length (minutes)	Immediate Predecessor(s)
A	0.5	-
B	0.7	A
C	1.5	A
D	0.3	C
E	0.6	B
F	0.2	C
G	0.4	E
H	0.3	D, F, G

- Draw and label a precedence diagram for the above tasks.
- What is the minimum cycle time
- Assuming the cycle time to be used for line-balancing is to be the minimum cycle time from part (b), develop a balanced line using the largest number of followers rule. Show the work stations, tasks available for assignment, tasks actually assigned, total task times assigned to work stations, and idle times.
- Determine the efficiency of the initial line balance based on part (c) above.

11. **(Service Facility Design)** A service organization has the following process design. There are two separate but identical service centers (with one employee each) that serve two distinct geographic areas. Each center has random service time with a mean of 10 minutes and a standard deviation of 10 minutes. Customers arrive at each center with an inter-arrival time of 12 minutes. The standard deviation of this inter-arrival time is 10 minutes.

Line 1**Line 2**

- What is the current average wait time in minutes?
- The organization is considering strategies to reduce wait times. In particular, it is considering reducing service time from 10 minutes to 5 minutes. In this case, the standard deviation of service time would also be reduced from 10 minutes to 5 minutes. What is the average wait time in minutes if this strategy is implemented?
- Another strategy to reduce waiting time involves standardizing the activity, which would reduce the standard deviation of the service time from 10 to 2.5 minutes. Service time of 10 minutes will remain unchanged. What is the average wait time if this strategy is implemented?
- The final strategy considered is to merge the two centers as shown below (in this case, the standard deviation of the inter-arrival times will be 6 minutes). What is the average wait time if this strategy is implemented?



Solution

1) (E)

2) (D)

Flow rate (FR) = $1500 / 15 = 100$ customers/ hour

Work-in-process (WIP) = 75 customers

FT = $WIP / FR = 75 * 15 / 1500 = \frac{3}{4}$ hr

3) (A)

WIP = \$45M

FR = 300 million dollars/ year

FT = $WIP / FR = 45 / 300 = 0.15$ year (or 54.75 days)4) (B) WIP = $FT / CT = (8+4) / 6 = 2$ customers

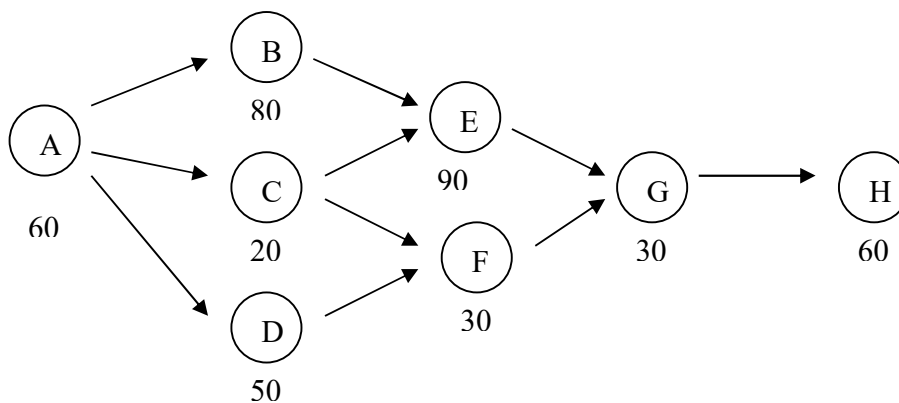
5) (E)

R = 255 per hour

I = 35

T = $35 / 255 = 0.137$ hours = 0.137×60 minutes = 8.24 minutes

6) (a)

(b) $CT = 8 * 3600 / 240 = 120$ seconds per unit

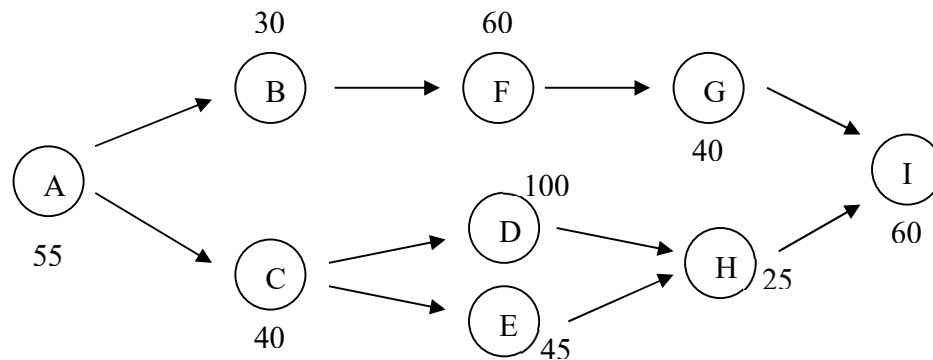
(c)

Workstation	Time Remaining	Eligible	Assign	Idle Time
1	120 60	A B, C, D	A D	10
2	120 40	B, C C	B C	20
3	120 30	E, F F	E F	0
4	120 90	G H	G H	30

(d) Efficiency = $T / N * C = 420 / 4(120) = 0.875$

7)

(a)

(b) Maximum production rate per day = $480 \times 60 / 100 = 288$

(c)

(i) Output rate with cycle time 3.0 minutes = $480 / 3 = 160$ units/day(ii) Cycle time leading to an output of 200 units per day = $480 / 200 = 2.4$ min = 144 s

(d)

(i) Total Weekly Operating Time = $3 \times 480 \times 5 = 7200$ min ≈ 432000 s.
CT = $43200 / 4320 = 100$ s.(ii) Theoretical minimum number of workstations, $N_{\min} = 455 / 100 = 4.55 \approx 5$ workstations

(e) Longest task time rule:

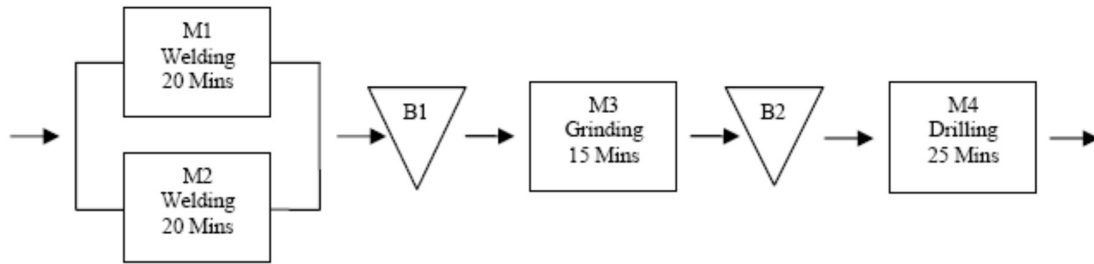
Work-station	Time Remaining	Eligible Task(s)	Assign	Idle
1	100 45	A B, C	A C	5
2	100	B, D, E	D	0
3	100 55 25	B, E B, H F, H	E B H	0
4	100 40	F G	F G	0
5	100	I	I	40

(f) For both heuristics,

$$\text{Efficiency} = \Sigma t_i / (N_a * C) = 455 / (5 * 100) = 0.91 \text{ or } 91\%$$

8) Stage 1 is the bottleneck with CT = 5 mins. Given that the average WIP is 3.2 units, based on Little's law, FT = WIP * CT = $3.2 * 5 = 16$.

9)
(a)



(b) The cycle time for the process is 25 mins per unit.

So capacity of the process = $60 * 1/25 = \underline{2.4 \text{ units/hour}}$

(c) Time it'll take to fill a rush order (FT) = $20 + 15 + 25 = 60 \text{ mins} = \underline{1 \text{ hour}}$.

(d) Initially, the bottleneck is the drilling stage, and thus we purchase a drilling machine.

Process CT after buying a drilling machine

= $\text{Max} \{20/2, 15, 25/2\} = \text{Max} \{10, 15, 12.5\} = 15 \text{ mins}$

Next, the grinding stage is the bottleneck, and we thus purchase a grinding machine.

Process CT after buying a grinding machine

= $\text{Max} \{20/2, 15/2, 25/2\} = \text{Max} \{10, 7.5, 12.5\} = 12.5 \text{ mins}$

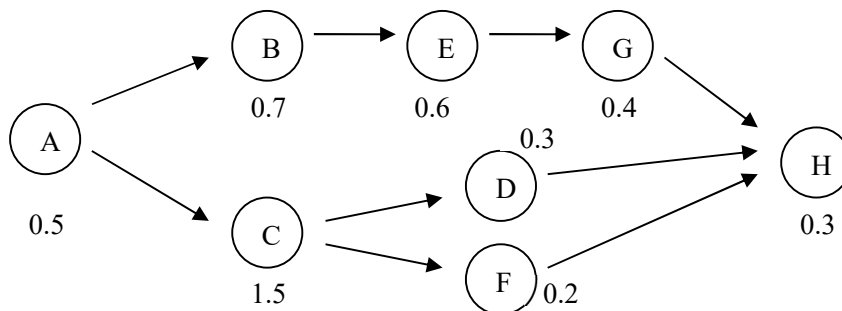
Now, the drilling stage is still a bottleneck, and we therefore buy another drilling machine.

Process CT after buying 2nd drilling machine

= $\text{Max} \{20/2, 15/2, 25/3\} = \text{Max} \{10, 7.5, 8.33\} = 10 \text{ mins}$

Hence the optimum (maximum) process CT is 10 minutes. The new capacity is 1/10 units per minute.

10)
(a)



(b) minimum cycle time = $1.5 \text{ min} = 90 \text{ s}$

(c)

Workstation	Time Remaining	Eligible	Assign	Idle
1	1.5	A	A	0.3
	1.0	B, C	B	
2	1.5	C, E	C	0
3	1.5	D, E, F	E	0
	0.9	D, F, G	G	
	0.5	D, F	D	
	0.2	F	F	
4	1.5	H	H	1.2

(d) Efficiency = $T / F = 4.5 / 6 = 0.75$ or 75%

11)

(a) $u = 10/12 = 5/6$

$$T_q = 10 \frac{5/6}{1/6} \frac{1}{2} \left(\left(\frac{10}{12} \right)^2 + 1 \right) = 42.36 \text{ min}$$

(b) $u = 5/12$

$$T_q = 5 \frac{5/12}{7/12} \frac{1}{2} \left(\left(\frac{10}{12} \right)^2 + 1 \right) = 3.026 \text{ min}$$

(c) $T_q = 10 \frac{5/6}{1/6} \frac{1}{2} \left(\left(\frac{10}{12} \right)^2 + \left(\frac{2.5}{10} \right)^2 \right) = 18.92 \text{ min}$

(d) $a = 6; \quad p = 10; \quad m = 2; \quad CV_A = CV_P = 1$

$$u = \frac{10}{6 \times 2} = 5/6$$

$$T_q = \frac{10}{2} \left(\left(\frac{5}{6} \right)^{\sqrt{6}-1} / \frac{1}{6} \right) \frac{1}{2} (2) = 23.03 \text{ min}$$